

# Ten Years of Event-Driven BGP Evolution in India and Bangladesh

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## Abstract

The global Internet's architecture is often shaped by local geography, economics, and geopolitics, creating complex dependencies between neighboring nations. This is particularly evident in South Asia, where the relationship between India and Bangladesh presents a compelling case of asymmetric interdependence. We conduct a ten-year (2015-2025) longitudinal and event-driven study of this relationship at the BGP control plane, using routing data from RIPE RIS. Our analysis reveals a deep-seated asymmetry: Bangladesh exhibits a structural and growing reliance on India for fundamental upstream resilience, while India deals with Bangladesh for strategic, commercially driven transit to its geographically isolated Northeastern region. We introduce the Relative Dependency Ratio (RDR), a metric to quantify the direction and magnitude of this cross-border transit. Using the RDR, we precisely measure the real-world impact of specific events such as the activation and suspension of a state-level transit agreement. Our findings provide a quantitative framework for understanding how national policy, commercial interests, and infrastructure fragility shape the Internet's evolution, offering critical insights for policymakers and network operators in emerging economies.

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## CCS Concepts

• **Social and professional topics** → **Transborder data flow;**  
**Governmental regulations;** • **Networks** → **Network measurement.**

## Keywords

Bangladesh, India, Border Gateway Protocol (BGP)

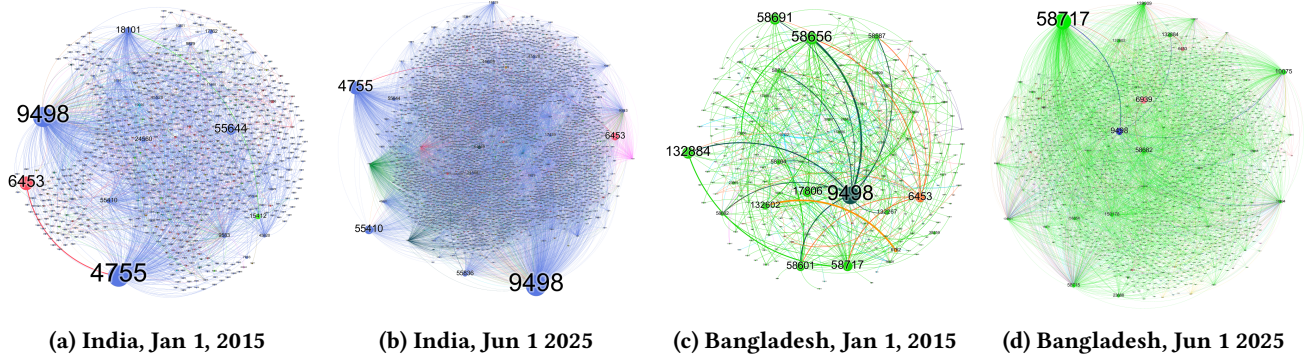
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## 1 Introduction

Although we often think of the Internet as a global, borderless network, its actual structure is shaped by real-world factors like national borders, geography, and economic relationships [9, 10]. In developing regions, these factors especially create complex dependencies between the countries, these dependencies then determine how resilient their networks are and how their traffic flows. Understanding how such relationships affect the Internet's performance, resilience, and control is a key challenge for both researchers and policymakers. This paper investigates one such critical relationship: the decade-long evolving connection between the networks of India, a regional powerhouse, and its smaller neighbor, Bangladesh.

The relationship between India and Bangladesh is a case of asymmetric connectivity. India has a massive population and diverse



**Figure 1: AS-level connectivity graphs for India (left pair) and Bangladesh (right pair) at the start (2015-01-01) and end (2025-06-01) of our study. Node size is proportional to prefix count; (left pair) blue edges denote internal peerings for India; (right pair) green edges indicate internal peerings for Bangladesh; other colored edges denote external upstream links.**

submarine cable connectivity [19], making it a dominant regional hub. Bangladesh, in contrast, has more limited international connectivity, with only two cable landings, forcing it to rely heavily on terrestrial fiber links to India [3]. This imbalance in infrastructure creates a natural dependency.

However, such reliance is not one-way. India’s geography creates a challenge: its landlocked Northeastern states can get faster and efficient Internet by routing through Bangladesh rather than across India’s difficult terrain [23]. This results in an interesting two-way reliance: Bangladesh depends on India for fundamental connectivity and resilience, while India depends on Bangladesh for a strategic and logistical shortcut. While this dynamic is generally known, its effects on the Internet’s control plane have not been systematically measured.

Douzet et al. [9] reported the intentional manipulation of traffic routing through BGP during a regional conflict. In addition to the study of BGP routing in a single-country context such as [17], the regulatory trends that restrict ISPs’ freedom to manage interconnection and routing in the context of China, Russia, and the USA have also been studied [11]. To the best of our knowledge, we present the first study of the influence of political and commercial events on the network infrastructure evolution of two neighboring countries. Through this work, we explore the following research questions:

**RQ1:** How has an asymmetric network dependency evolved alongside the growth of each country’s network?

**RQ2:** How do commercial deals and political tensions appear in BGP routing data?

**RQ3:** How does the network dependency hold up during major disruptions, like infrastructure failure?

Our work makes the following contributions:

1. We characterize the ten-year growth of India’s and Bangladesh’s AS-level networks, showing how India’s connectivity remains centralized around a few large providers, whereas Bangladesh’s centralization stems from its limited transit diversity and partly from its dependence on India.

2. We introduce a metric, the Relative Dependency Ratio (RDR), to quantify the direction and strength of their mutual transit dependency. Using this, we capture the fluctuating impact of the transit agreement between the countries’ state-owned ISPs.

3. Through case studies of key events, such as the SMW-5 submarine cable outage and the launch of Reliance Jio, we show how infrastructure disruptions and policy interventions shape this relationship.

This paper is organized as follows. Section 2 reviews related work on measuring internet resilience and geopolitics. Section 3 details our data and methodology. Section 4 presents our longitudinal analysis of network growth, while Section 5 uses our event-driven case studies to understand the key events that have defined this asymmetric interdependence. We then discuss the limitations of this study and further steps in Section 6 before concluding.

## 2 Motivation & Background

A growing body of work investigates how resilience and redundancy manifest at national and regional scales. Carisimo et al. [8] examine Venezuela’s control-plane topology during a political and economic crisis, showing how AS-path churn and upstream loss can serve as indirect indicators of infrastructure fragility. Similarly, Opalinski et al. [14] conceptualise resilience in constrained geopolitical contexts, emphasising the importance of diverse upstream connectivity and state infrastructure policy in sustaining access and posing over-centralization of international egress as a risk to national connectivity. APNIC’s 2023 commentary [4] reinforces this view, highlighting how emerging markets that rely heavily on a small number of upstream providers or submarine cable landing points face heightened vulnerability to physical disruptions and policy-level changes. Our work builds upon these concerns and extends the discussion to a regional, event-driven context by empirically examining how infrastructure fragility and resilience have evolved in Bangladesh and India over the past decade.

While both nations have experienced significant AS-level growth over the past decade, their external connectivity profiles reveal pronounced asymmetries. In its August 2023 regional assessment [3] APNIC highlights Bangladesh’s increasing reliance on external

upstream providers, such as Hurricane Electric, alongside India’s persistent dependence on a narrow set of ASes, as indicative of a broader deficit in regional path diversity.

A complementary stream of political-economic scholarship has shown that the very same structural attributes we measure for resilience, such as path diversity, upstream churn, and landing station redundancy, are often the product of commercial and state strategies [9, 10, 12]. These studies imply that resilience metrics cannot be interpreted in isolation from the geopolitical and commercial contexts that shape underlying infrastructure decisions.

### 3 Data and Methodology

#### 3.1 Datasets

We use publicly available control-plane BGP data from the RIPE Routing Information Service (RIS) [16]. Specifically, we extract complete RIB dumps captured on the first day of each month from January 2015 through June 2025, resulting in 126 monthly snapshots per country. Each snapshot includes all globally visible IPv4 and IPv6 prefixes along with their AS-path attributes. This monthly granularity balances temporal resolution with robustness to short-lived anomalies such as route flaps or misconfigurations, allowing us to focus on long-term structural trends. To identify routing and interconnection patterns specific to India and Bangladesh, we filter AS-paths by origin ASes located in each country using WHOIS and CAIDA datasets.

Additionally, we use metadata from TeleGeography’s Submarine Cable Map [19] to track international cable deployments, landing points, and connectivity timelines, which we correlate with observed routing changes during specific events.

Finally, we use CAIDA’s AS relationship dataset [7] to distinguish between peering and transit relationships between different Autonomous Systems.

#### 3.2 Analysis Methods

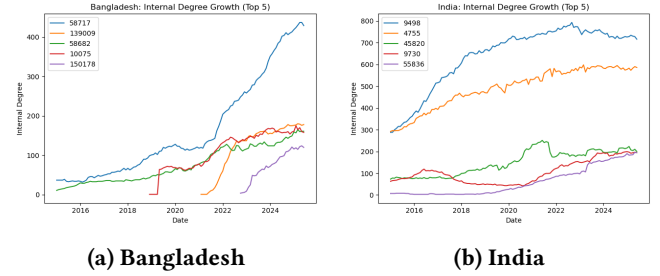
Our analysis focuses on how AS-level connectivity evolves, using several key metrics. These include internal degree, which measures the number of domestic AS neighbors per AS; upstream diversity, referring to the number of distinct foreign upstreams; and address-space distribution, which tracks the share of total IPv4 and IPv6 space held by each AS. We also introduce the Relative Dependency Ratio (RDR) to measure the directional balance of cross-border transit between countries. We apply these metrics over time-series data obtained from RIPE RIS to assess the impact of policy changes, infrastructure disruptions, and commercial agreements on routing behavior.

### 4 Longitudinal Study

Over the decade from January 2015 to June 2025, India and Bangladesh saw a dramatic expansion of their Autonomous System (AS) topologies. As seen in Table 1, India’s internal ASes more than quadrupled and external providers doubled, while Bangladesh’s internal ASes grew seven-fold and external about four-fold. Figure 1(a,c) captures the January 2015 view: India, dominated by Bharti Airtel (AS 9498) and TATA (AS 4755) with heavy reliance on TATA US (AS 6453) as an external provider; for Bangladesh, a collection of small networks largely dependent on Airtel. By June 2025 (Figure 1(b,d)), India’s

| Year     | India    |          | Bangladesh |          |
|----------|----------|----------|------------|----------|
|          | Internal | External | Internal   | External |
| Jan 2015 | 677      | 193      | 208        | 58       |
| Jun 2025 | 2774     | 428      | 1442       | 236      |

**Table 1: Number of distinct internal (to each country) and external ASes (immediate foreign ASes that internal ones peer with) in India and Bangladesh in Jan 2015 vs. Jun 2025.**



**Figure 2: Internal degree growth for the top 5 ASes for India and Bangladesh**

control-plane remained centered on the same handful of nodes even as new players emerged, and Bangladesh’s network grew around Summit Communications (AS 58717), with the presence of a new external provider, Hurricane Electric (AS 6939), becoming prominent.

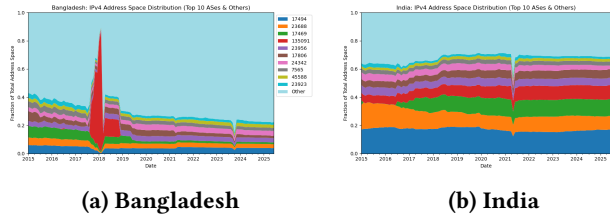
In this section, we chart longitudinal trends along three dimensions: internal network, IPv6 adoption, and external connectivity. Section 5 then leverages an event-driven framework to link connect observed inflections to major policy, infrastructure, and commercial events.

#### 4.1 Internal Network & IPv4 Distribution

A country’s internal network growth, measured by increased ASes and peering links, does not inherently indicate meaningful infrastructure advancement. To assess such growth, we examine the internal degree of ASes in each country for the extent of intra-connectivity and IPv4 space distribution to measure an AS’s potential network footprint.

**4.1.1 Internal Connectivity.** To gauge each domestic network’s mesh density, we track internal degree, which is defined as the number of same-country neighbors for each AS. We are interested in the growth of the most well-connected ASes today. Hence, in Figure 2, we focus on the top 5 ASes in India and Bangladesh by internal degree in June 2025, and plot their evolution since 2015.

For Bangladesh, Figure 2(a) depicts Summit (AS 58717) showing rapid growth in internal connectivity compared to the rest of the ISPs due to its operation of the largest fiber optic network inside the country and its role as a licensed International Internet Gateway (IIG) and Terrestrial Cable (ITC) operator under Bangladesh’s 2010 ITC and IIG licensing regime (Section 5.2.1). Other Bangladeshi ISPs show more modest, but steady, growth in internal degree, reflecting the gradual expansion of their peering relationships and network interconnections over time.



**Figure 3: Fraction of total IPv4 address-space held by the on-average top-10 ASes and all others in Bangladesh and India.**

For India, Figure 2(b) shows that Bharti Airtel (AS 9498) and TATA (AS 4755) are the most internally well-connected, due to their long-standing roles as backbone providers with widespread infrastructure and early involvement in domestic peering. Reliance Jio (AS 55836) began to ramp up post-2018, as they entered the Indian market in 2016 with native IPv6 deployment, discussed later on in Section 5.1.2.

**4.1.2 IPv4 Address-Space Distribution.** As the dominant protocol, IPv4 underpins the majority of routed Internet traffic and addresses. How that address space is distributed across ASes reveals the potential number of end-hosts and networks it can directly reach, and the potential resistance of the national Internet to individual failures.

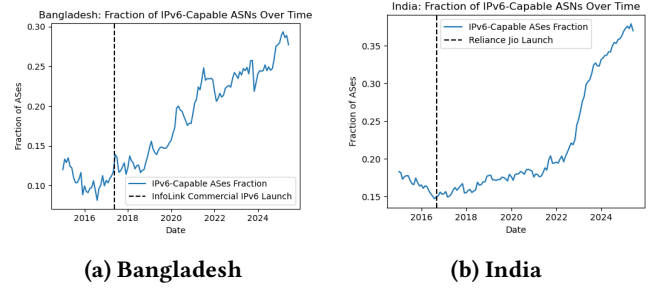
Figure 3 shows the fraction of total IPv4 address space held by the on-average top-10 ASes (those ASes that appear in the top ten most consistently over the decade) in each country versus all others over time. In Bangladesh (Figure 3(a)), the "Other" category consistently accounts for more than half the space, with the top-10's share showing a gradual decline over time. This long-tail distribution implies that a wide range of smaller networks hold a share in the address space. This suggests that an outage affecting a single network is less likely to threaten nationwide reachability.

By contrast, India (Figure 3(b)) exhibits a more concentrated distribution. Three providers, Bharti Airtel (AS 9498), Bharti Airtel GPRS (AS 45609), and Reliance Jio (AS 55836), account for about 40% of India's IPv4 space by 2025. An incident affecting Airtel or Jio would significantly impact the country's addressable Internet.

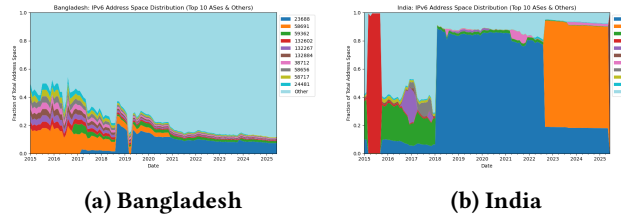
## 4.2 IPv6 Adoption

IPv6 adoption stands out as a strong indicator of modern network infrastructure. Unlike IPv4, deploying IPv6 often requires substantial hardware, software, and operational upgrades. As a result, sustained growth in IPv6 prefix announcements typically reflects deliberate infrastructure investment and future-oriented network planning by operators.

We measure IPv6 prevalence at the BGP level to gauge operator deployment. Figure 4 shows the fraction of active IPv6-capable ASes in each country for each month over the decade. India's (Figure 4(b)) fraction plateaus around late 2016, experiencing an increase in IPv6 adoption after Reliance Jio's IPv6 launch (Section 5.1.2). Bangladesh (Figure 4(a)) starts at 15% adoption and doubles it by 2025. Compared to India's early IPv6 roadmap (Section 5.1.1), Bangladesh had no such early IPv6 initiatives and would not obtain commercial IPv6 availability until InfoLink's IPv6 launch in early 2017.



**Figure 4: Fraction of IPv6 Capable ASes over time in Bangladesh (a) and India (b)**



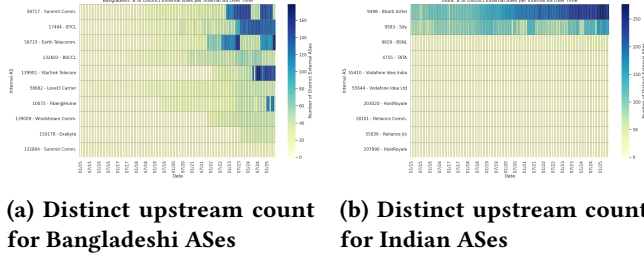
**Figure 5: Fraction of total IPv6 address-space held by the on average top-10 ASes and by all others in Bangladesh and India.**

To capture the distribution of IPv6 allocations within a country, we break down each country's total IPv6 address space among its on-average top-10 IPv6 announcing ASes. Figure 5 shows how each country's total IPv6 address space is split between the top-10 ASes and all others.

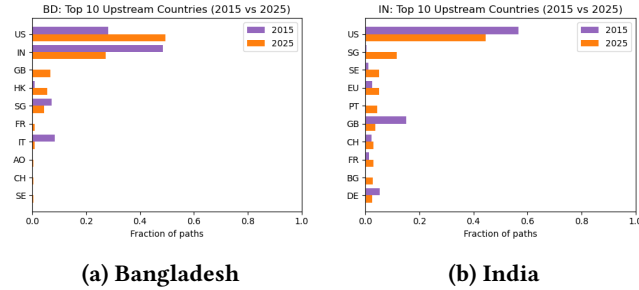
In Bangladesh, Figure 5a, early IPv6 presence, although small, was from a handful of ASes. This pattern is primarily due to ASes testing IPv6 through the Bangladesh Internet Exchange (BDIX), while very few initially had dual-stack capability [1]. From 2020 onwards, as other networks began gradually adopting IPv6 on a small scale, the top 10 ASes' share of IPv6 usage shows a steady decline. India's (Figure 5b) address space is far more concentrated on a handful of providers. The most pronounced and sustained inflection occurred in 2018, a few years after Reliance Jio's all-IPv6 4G launch, owning roughly 80% of the IPv6 address space till late 2022. In late 2022, India's Public Anycast DNS (AS 148000) surged to hold the largest addressable space. This growth is discussed further in Section 5.1.2.

## 4.3 External Connectivity

External connectivity shapes a country's resilience by determining how traffic exits to the global Internet. Greater upstream diversity can improve redundancy, though multiple peers may still share the same physical infrastructure, limiting true resilience. Conversely, dependence on a few upstream ASes or countries increases vulnerability to outages and policy shifts. We examine this evolution for India and Bangladesh through (1) the number of distinct upstream providers per AS and (2) the distribution of BGP paths across upstream countries and ASes.



**Figure 6: Number of distinct external upstream ASes for the top 10 internal ASes in Bangladesh and India (Jan 2015 to June 2025).**

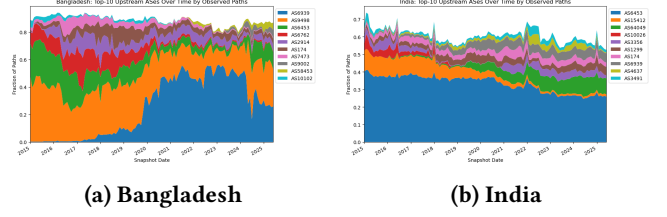


**Figure 7: Fraction of BGP paths via the top-10 upstream countries in 2015 (purple) and 2025 (orange).**

**4.3.1 Number of Distinct Upstreams.** The resilience of a national network depends on how traffic leaves the country. We look at the distinct number of external upstream providers a given AS in a country peers with. Figure 6(a) shows that in Bangladesh, the leading ASes had few external peers until 2021, after which AS 58717 and AS 10075 increased their number of distinct external upstreams to over 150, showing increased diversification in upstream transit. For India, we see in Figure 6(b) that Bharti Airtel (AS 4998) steadily increased its number of distinct upstream connections from around 100 to 270. Interestingly, AS 4755, TATA Communications India, a significant service provider in India, remains relatively stagnant in upstream diversification due to its heavy reliance on TATA Communications America.

**4.3.2 External AS distribution.** We next examine how each country’s external transit dependency has evolved in terms of the countries providing upstream connectivity and the specific ASes involved.

Figures 7a and 7b compare the fraction of BGP paths routed via the top ten upstream countries in January 2015 versus June 2025. In 2015, Bangladesh’s transit was primarily routed through India (accounting for approximately 45% of observed paths) and the United States (representing 35% of the observed paths), with smaller proportions transiting via Hong Kong, Singapore, and Italy. By 2025, India’s share decreased to about 30%, while the United States’ share increased to over 50%. This shift reflects a move away from regionalized connectivity. India’s upstream connectivity has persistently depended on the US, with Singapore and European providers holding substantially smaller shares.



**Figure 8: Evolution of the fraction of paths through the top-10 upstream ASes.**

Figures 8a and 8b show the time-series evolution of the fraction of paths passing through the top-10 upstream ASes, where the top-10 is selected based on today’s BGP graph. At the AS level, Bangladesh in 2015 relied heavily on AS 9498 (Bharti Airtel) and AS 6453 (Tata Communications US), with minor contributions from AS 6939 (Hurricane Electric) and AS 174 (Cogent). Around 2020, AS 6939’s share surges to over 50%, making Hurricane Electric the primary Bangladeshi transit provider, while AS 9498’s share reduces. This reaffirms Bangladesh’s shift away from dependence on regional transit. On the other hand, India’s upstream dependence has remained relatively stable, having primarily relied on AS 6453. Combined with the previous egress diversity, this underscores India’s persistent centralized transit market compared to Bangladesh.

#### 4.4 Cross-Border Dependence

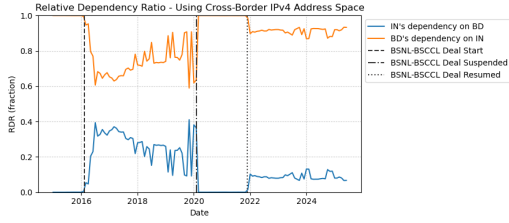
Bangladesh and India share a long-standing and highly asymmetrical interdependence for international transit. With only two submarine cables landing directly in Bangladesh [19], the country leans on terrestrial fiber links into India for resilience. Conversely, India’s northeastern states face geographic constraints in connectivity to mainland India, preferring to route traffic through Bangladesh to improve latency. To quantify this bilateral reliance, we examine the balance of IPv4 address space carried across the border over time.

We capture this with the Relative Dependency Ratio (RDR), defined for country A on country B as:

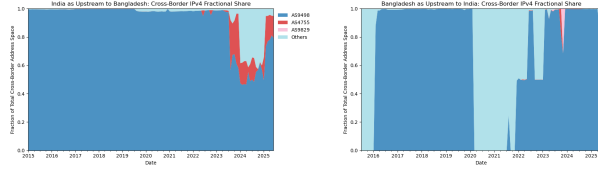
$$RDR_{A \rightarrow B} = \frac{IP_{A \rightarrow B}}{IP_{A \rightarrow B} + IP_{B \rightarrow A}}$$

Where  $IP_{A \rightarrow B}$  is the total IPv4 address space of prefixes originating in A whose immediate upstream AS lies in B. An RDR near one means that majority cross-border address space originates from A towards B, and nearly none the other way. An RDR near zero indicates the reverse. We computed RDR for IPv4 and IPv6 separately. For IPv6, the  $RDR_{BD \rightarrow IN}$  consistently holds a value near 1, with no notable changes around the events discussed later. For the analysis performed moving forward, we restrict it to IPv4 in order to better understand the shifts in cross-border address space corresponding to the events discussed.

Figure 9 illustrates the symmetric RDR trends for Bangladesh on India (orange) and India on Bangladesh (blue). Before February 2016, India’s  $RDR_{IN \rightarrow BD}$  is near zero (symmetrically, Bangladesh’s  $RDR_{BD \rightarrow IN}$  is near 1), reflecting the negligible use of Bangladeshi upstreams for transiting Indian traffic. Moreover, throughout this period, Bangladesh has maintained a relatively high dependency



**Figure 9: RDR between India and Bangladesh represents the ratio of cross-border IPv4 address space**



**(a) Indian ASes as upstream (b) Bangladeshi ASes as upstream to Bangladesh**

**Figure 10: Fraction of Bangladesh's cross-border address space transited by ASes in India (a) and vice versa (b)**

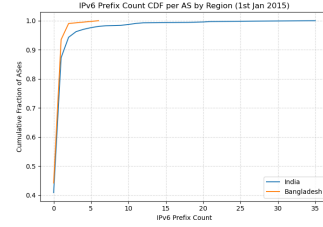
on India compared to the reverse. The early 2015, early 2020, and late 2021 inflection points are related to the bandwidth agreement between India's state-owned Bharat Sanchar Nigam Limited (BSNL) and Bangladesh's state-owned Submarine Cable Company (BSCCL). These inflection points are investigated via a more granular view of RDR and upstream changes for BSNL in Section 5.3.

The inter-dependency between India and Bangladesh has been shaped by a few prominent players, such as the state-owned BSNL in India and BSCCL in Bangladesh. In Figure 10, we plot the cross-country IPv4 address space passing through a select few ASes in each country. For the address space originating from Bangladesh and passing through India in Figure 10(a), we see that Bharti Airtel (AS 9498) carries over the most significant fraction of the address space due to its prevalence as an upstream for Bangladesh. Similarly, TATA (AS 4755) recently started playing a moderate role.

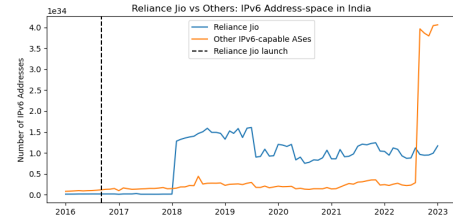
On the other hand, for the Indian origin address space crossing over into Bangladesh, in Figure 10(b), we see that from 2016 to 2020, and late 2021 onwards, BSCCL is responsible for the majority of cross-border Indian address space. This coincides with the inflection points discussed earlier in India's  $RDR_{IN \rightarrow BD}$  value as seen in Figure 9. Moreover, although a few other ASes in Bangladesh are observed to carry Indian address space, Figure 10(b) also shows how negligible this space is compared to the address space BSCCL transits.

## 5 Events & Impacts

In this section, we first give an overview of the general policies and events that shaped the IPv6 landscape we see in the two countries today. We then examine key events that influence the cross-border dependence between the two countries.



**Figure 11: Empirical CDF of number of IPv6 prefixes per AS in Bangladesh and India**



**Figure 12: Monthly IPv6 address-space announced by Jio vs. all other Indian ASes. (Vertical dashed line marks 5 Sep 2016 launch.)**

## 5.1 IPv6 Reforms & Developments

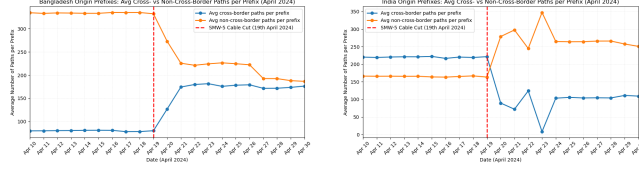
**5.1.1 India's IPv6 Deployment Roadmap (June 2010).** In June 2010, India's Department of Telecommunications issued a National IPv6 Deployment Roadmap, requiring all major ISPs to support IPv6 by the end of 2011 and government networks to adopt IPv6 by March 2012.

Figure 11 shows the empirical CDF of IPv6 prefix counts per AS on 1st January 2015 for India (blue) and Bangladesh (orange). Both have the majority of their IPv6-capable ASes announcing five or fewer prefixes. However, we observe a long tail for India of a handful of operators announcing many IPv6 prefixes. This disparity reflects India's 2010 IPv6 roadmap, driving multi-block IPv6 rollouts via a few large providers, compared to Bangladesh's minimal initial deployments.

**5.1.2 Reliance Jio IPv6 Launch in India (Sept 2016).** In September 2016, Reliance Jio (AS 55836) entered India's market as the first IPv6-only 4G mobile network. Jio received its initial allocation from APNIC in 2010 [6], and ran a nine-month trial period on its IPv6 network internally before releasing to the public [2].

Figure 12 overlays Jio's announced IPv6 address space against the aggregate of all other IPv6-capable Indian ASes. We observe in the February 2018 snapshot an increase in Jio's IPv6 space. Jio shifted from announcing a handful of subnets from its 2010 APNIC allocated /29 IPv6 block, to announcing a 2017 allocated /22 block [6]. As observed in Figure 5(b), Jio's share of the IPv6 address space jumped from around 10% to nearly 90% in 2018.

Reliance Jio's dominance of the BGP-visible IPv6 space lasted until late 2022, when AS 148000 (Indian Public DNS) surged to owning around 80% of the visible space. In October 2021, APNIC allocated a /20 block to AS 148000, which we observe 148000 announce in the September 2022 snapshot, causing the observed burst [5].



(a) For Bangladeshi origin prefixes with at least one cross-border path (b) For Indian origin prefixes with at least one cross-border path

Figure 13: Average number of cross-border vs non-cross-border paths per prefix for (a) Bangladesh, and (b) India

## 5.2 Bangladesh's Dependence on India

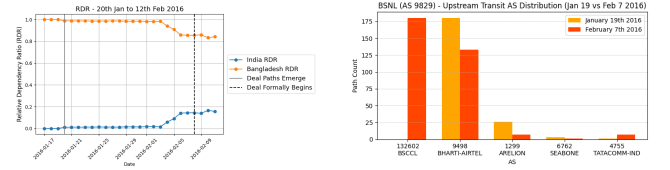
**5.2.1 Bangladesh Licensing of IIG/ITC Operators (April 2012).** In 2010, the Bangladesh Telecommunications Regulatory Commission (BTRC) introduced the International Long-Distance Telecommunications Service (ILDTS) licensing regime. Before this, Bangladesh's only international connectivity was the SEA-ME-WE-4 submarine cable. Under ILDTS, they began awarding licenses to private International Internet Gateway (IIG) and International Terrestrial Cable (ITC) operators, with the first six licenses issued by April 2012. This liberalization aimed to break the single-cable dependency by enabling multiple terrestrial crossings into India and improving resilience.

**5.2.2 SMW-5 Malacca Strait Outage (April 2024).** On 19th April 2024, a fault in the Malacca Strait severed the SEA-ME-WE 5 (SMW-5) cable, cutting Bangladesh's primary submarine cable link to Southeast Asia.

To study Bangladesh's rerouting around the cable cut, we focus on India's role. We filter out Bangladeshi-origin prefixes that crossed into India at least once from 16th April to 24th April. As a prefix can have multiple unique AS paths, we are interested in the average number of paths per Bangladeshi-origin prefixes that cross over into India versus those that have a non-Indian external connectivity. Figure 13(a) plots the average number of distinct cross-border and non-cross-border paths seen per prefixes originating from Bangladesh that cross over India at least once. Before the cut, such prefixes traversed around 80 distinct cross-border paths and 330 non-cross-border paths. Immediately after the outage, the average number of paths per Bangladeshi-origin prefixes crossing over into India more than doubled. In contrast, non-cross-border paths decreased by nearly the same amount, showing a shift of non-cross-border paths to now go through India.

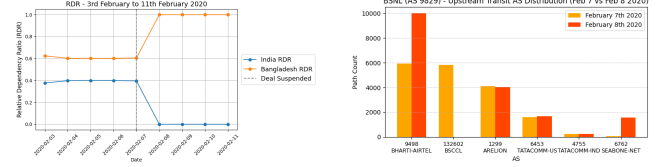
Figure 13(b) shows the opposite case, the distribution of the average number of paths per Indian origin prefixes. Those routed via Bangladesh see the paths shift from transiting via Bangladesh to a non-Bangladeshi external upstream. Hence, due to the cable outage, India rerouted its paths away from Bangladesh through alternative upstreams.

These results demonstrate that the SMW-5 outage triggered widespread control-plane churn, specifically in Bangladesh: cross-border paths increased via the terrestrial links, highlighting the importance of terrestrial connectivity to India for Bangladesh.



(a) RDR over the 2016 deal launch (day-level granularity) (b) Upstream ASes to BSNL before and after the Feb 2016 deal

Figure 14: Relative Dependency Ratio (RDR) and changes in upstream to BSNL (AS 9829) around the Feb 2016 deal



(a) Relative Dependency Ratio over the deal suspension (day-level granularity) (b) Upstream ASes to 9829 before and after the deal suspension

Figure 15: Relative Dependency Ratio (RDR) and changes in upstream to BSNL (AS 9829) around the Feb 2020 deal suspension

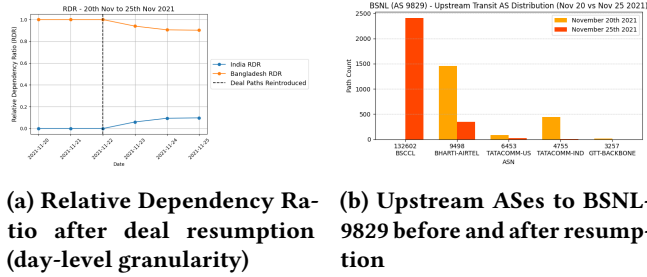
## 5.3 India's Dependence on Bangladesh

In this section, we focus on the historical bandwidth trade agreements between India and Bangladesh to provide transit for the northeastern Indian states.

**5.3.1 BSNL-BSCCL Transit Deal Start (June 2015).** In June 2015, the two state-owned telecom operators BSNL (AS 9829) and BSCCL (AS 132602) of India and Bangladesh, respectively, reached an agreement to enable bandwidth export from Bangladesh to India [13]. The agreement established a three-year lease of international transit for BSNL through BSCCL. The agreement officially came into force on 8th February 2016. Before this, India had an  $RDR_{IN \rightarrow BD}$  score of near zero. After the deal, India's score rises to near 0.2 over the next few days, as seen in Figure 14a, eventually rising to 0.4 over the next few months (Figure 9).

Before the deal, BSNL's upstreams for the affected prefixes were primarily routed through providers such as Bharti Airtel (9498) and Arelion (1299), as shown in Figure 14(b). We observe the initial emergence of Bangladesh as an upstream beginning around January 20th, 2016. By February 8th, following the formal commencement of the deal, BSCCL had become the dominant upstream, with 180 observed paths, marking the start of a prominent international upstream for India.

**5.3.2 BSNL-BSCCL Suspension (Feb 2020).** On February 7th, 2020, after four years of the deals beginning, BSNL stopped purchasing bandwidth from BSCCL due to an inability to meet payments [22, 26]. We observe this transition through RDR as shown in Figure 15(a). In the days leading up to the suspension, India also showed



**Figure 16: Control-plane routing changes following the November 2021 deal resumption**

a moderate RDR of nearly 0.4, indicating an almost equivalent interdependence between the two countries. However, immediately following the deal's suspension on February 7th, the RDR for India dropped to near 0. Only one IPv4 prefix originating from India passed through Bangladesh, alongside a handful of IPv6 prefixes, which we discount when calculating RDR.

To understand the routing adjustments post-suspension, we observe the set of prefixes containing at least one path passing through BSNL with BSCCL as its upstream before the suspension, and how such paths shifted from BSCCL as an upstream to a different transit upstream. Looking at the new upstream distribution in Figure 14(b), we see the vacuum left by BSCCL gets filled by Bharti Airtel (AS 9498), which appears with double the number of paths than before. Such a suspension indicates that northeast India must route via the longer physical path through India once again.

**5.3.3 BSNL-BSCCL Deal Resumption (Nov 2021).** According to an official report by the Ministry of Communications, India, the transit deal between BSNL and BSCCL formally resumed on November 26, 2021 [15, 28]. We observe the reappearance of BSCCL as an upstream for BSNL initially on 22nd November 2021. Examining the RDR in Figure 16(a) from 20th to 25th November in 2021, we observe that while Bangladesh continues to predominantly rely on India for upstream transit, there is a notable shift as BSNL begins to route through BSCCL once again.

As shown in Figure 16(b), the resumption led to a substantial reconfiguration of routing dependencies. Bharti Airtel (AS9498), which had previously filled in for the removal of BSCCL, saw a sharp drop in its transit role. In contrast, BSCCL (AS 132602) rapidly regained its position as a dominant upstream for BSNL, showing a return to Bangladesh-based transit for the Northeastern Indian states.

**5.3.4 Private Companies Transit Proposal (March 2023).** In March 2023, Summit Communications (AS 58717) formally submitted a proposal to BTRC for providing transit to Northeastern Indian states in agreement with Bharti Airtel (AS 9498) in India [25]. Around the same time, Fiber@Home (AS 10075) advanced a similar plan. While both initiatives gained early momentum, BTRC revoked its prior recommendation in December 2024, citing strategic concerns [23, 27]. We never see Fiber@Home (AS 10075) or Summit (AS 58717) act as upstream to Bharti, as the agreement was never officially approved by BTRC. In Figure 10, we observe the presence of Summit and Fiber@Home as upstream ASes to India; these correspond to

other ASes that they are upstream of. The spike is likely a transient network event.

## 6 Discussion

Our work underscores broader implications for emerging networks and their resilience, highlighting how market-driven growth does not necessarily translate into robustness. Both countries display centralization of external connectivity. Bangladesh's dependence on India for transit and India's on TATA's global network highlight that commercial and regional powers highly influence network shape. The RDR and its application to case studies provides a quantitative framework for understanding how geopolitical decisions manifest in routing behaviour and how politically mediated agreements can rapidly reshape cross-border dependencies.

Simultaneously, there are limitations to our approach. Our reliance on BGP control-plane data means we observe routing advertisements rather than actual traffic flows. Our analysis captures the structure of possible paths but cannot confirm whether, or how frequently, they are used. Additionally, while we correlate many observed changes with known infrastructure events or policy agreements, our ability to explain operator behavior is constrained. The impact of notable events such as the Covid-19 pandemic can not be easily correlated with shifts in routing behaviour without explicit insight into the operator's motivation behind any such changes.

Future work could address these gaps through a combination of data plane measurements and field work. Measurements such as traceroutes and ping-based latency probes can verify whether advertised BGP paths are traversed and identify alternate routes taken during network events. Finally, while this study focuses on India and Bangladesh, the framework and methodology can be applied to other regions with similar characteristics. Looking at other developing regions could offer comparative insights into how geopolitical and commercial forces interact to shape regional internet resilience.

## 7 Conclusion

Over the past decade, India and Bangladesh have developed a tightly coupled but uneven Internet relationship. Bangladesh relies on Indian upstreams for international reach, while India uses Bangladesh as a solution for its Northeastern region's connectivity challenges. This interdependence, though asymmetrical, shows us how strategic and economic considerations often take precedence over models that would promote greater resilience. Our findings highlight the need for deliberate policy and commercial planning in emerging economies to mitigate routing fragility and ensure resilience in the face of internet events. As the Internet continues to evolve, the interplay of technical and non-technical forces will remain central in shaping how countries connect to the global Internet.

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